

DRISHYAM

From Footage to Action

AI Evaluation & Cost Document

Accuracy Targets, Test Cases, Cost Control

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1. Accuracy Requirements (MVP Targets)

- Weapon detection: high recall preferred (minimize missed weapons); precision after temporal filter ≥ 0.85 in pilot environments.
- Violence: balanced precision/recall; prioritize reduction of obvious false positives (sports, play).
- Intrusion / Loitering: high precision once zones and thresholds are tuned per site.
- Overall false positive rate for operator-facing alerts $< 10\%$ after initial tuning.

2. Evaluation Protocol

Maintain a golden set of labeled clips (positive + hard negative). Run offline evaluation on every model candidate. Record precision, recall, F1, latency, and confusion examples. Human review of FP/FN samples before promotion to production.

3. Hallucination / Bad Output Handling

AI does not generate free text in MVP; outputs are structured classes + scores + boxes. “Bad outputs” = wrong class or spurious detection. Mitigated by temporal filters, confidence thresholds, and operator FP feedback.

4. Cost Drivers & Control

- Dominant cost: edge hardware (CapEx) or cloud GPU (OpEx) if used heavily.
- Prefer edge inference → predictable cost per camera, better privacy.
- Analyze at reduced FPS/resolution where acceptable.
- Clip storage: short clips only; aggressive retention; compressed codec.
- Cloud fallback: rate-limited and used only when edge unavailable or for rare heavy models.

5. Rough Cost Guidance (Order of Magnitude)

Edge node capable of 8–16 streams: hardware cost amortized over years. Cloud GPU inference if used 24/7 for many streams becomes expensive quickly — hence edge-first strategy. Object storage for clips is relatively cheap if retention is short.